Mellanie Rodriguez

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. Electrical Engineering and Computer Science, Minor in Design — *Expected May 2029*
Coursework: Circuits and Electronics (6.200), C and Assembly (6.1903), Discrete Math (6.1200), E&M (8.02), Classical Mechanics (8.01), Multivariable Calculus (18.02), PCB Design (6.S089)

Experience

Incoming Firmware Undergraduate Intern — October 2026
Auto-ID Labs @ MIT — *Cambridge, MA*

Mechatronics Engineering Researcher — Jun. 2026 – Sep. 2026
MIT Media Lab — *Cambridge, MA*
- Designed a robotic electrospinning end effector in Onshape and Fusion 360, cutting electric field arcing by 10%
- Built an ESP32 and NEMA 17 stepper motor fluid delivery system for automated polymer dispensing
- Wrote Python tools to analyze electric field distributions and guide high-voltage needle placement

Manufacturing Intern — Apr. 2025 – May 2025
Inkbit — *Medford, MA*
- Reverse engineered damaged aircraft parts through CAD reconstruction and ran CFD to improve the design

Mechanical Engineering Intern — Jun. 2024 – Aug. 2024
MIT Lincoln Laboratory — *Lexington, MA*
- Designed a self-playing xylophone in SolidWorks that plays "Mary Had a Little Lamb"
- Used Young's modulus and density to choose Aluminum 6061 and predict each bar's resonant frequency
- Tuned bar length and thickness to hit target pitches by relating stiffness, mass, and frequency

Robotics Intern — Jul. 2023
Tufts Engineering Design Lab — *Medford, MA*
- Designed an autonomous recycling robot using a Raspberry Pi 3B+ and supervised machine learning

Projects

FM Radio Receiver PCB | *Altium, LTspice* — Jan. 2026 – Present
- Designed schematic and PCB layout for an FM receiver with an RF signal path and audio amplifier stage
- Simulated amplifier gain and filter behavior in LTspice to select capacitor and inductor values

Smart Beverage Frother | *ESP32, C++, I2C* — Dec. 2025 – Present
- Built ESP32 control for temperature regulation and motor driving, with a TFT display and potentiometer input
- Designed custom parts for Formlabs SLA/SLS printing and laser cut metal

Walking Robot | *Arduino Nano, Raspberry Pi 4, Python* — Jun. 2025 – Jul. 2025
- Built a two-layer control system with coordinated servo control and OpenAI API conversation

Activities

Project Mentor — Dec. 2025 – Present
MIT Edgerton Center and Metropolis — *Cambridge, MA*
- Mentor 100+ students per semester at two makerspaces
- Teach power supplies, laser cutting, FDM and SLA fabrication basics, and iterating on projects

Member — Sep. 2025 – Present
Society of Hispanic Professional Engineers (SHPE) — *Cambridge, MA*
- Attend chapter events and volunteer in STEM classes for 20+ Hispanic students in Boston Public Schools

Skills and Honors

Languages: Python, C/C++

Hardware: ESP32, Arduino, Raspberry Pi, I2C

Tools: Altium, LTspice, SolidWorks, Fusion 360, Onshape, Git, PlatformIO

Honors: Amazon Future Engineer Scholar, Microsoft Disability Scholar, U.S. Presidential Scholar Nominee, 2nd Place MakeMIT, 1st Place Design Museum Everywhere